WEB DEV WITH REACT

USEEFFECT & APIS

LIVE VS
STATIC WEATHER

WHERE DOES THE LIVE DATA COME FROM?

How does a weather app show live updates? One app shows the same temperature all day — that's static! Another updates every minute — that's live! Think about Swiggy or Zomato: they always show your order status in real time. How do they do it?

CAN YOU SPOT THE DIFFERENCE? 🌤️

STATIC VS LIVE: WHAT'S THE DIFFERENCE?

A static app shows old data — like a photo! A live app fetches fresh data every time — like a video call! Live apps talk to something called an API to get new information instantly.

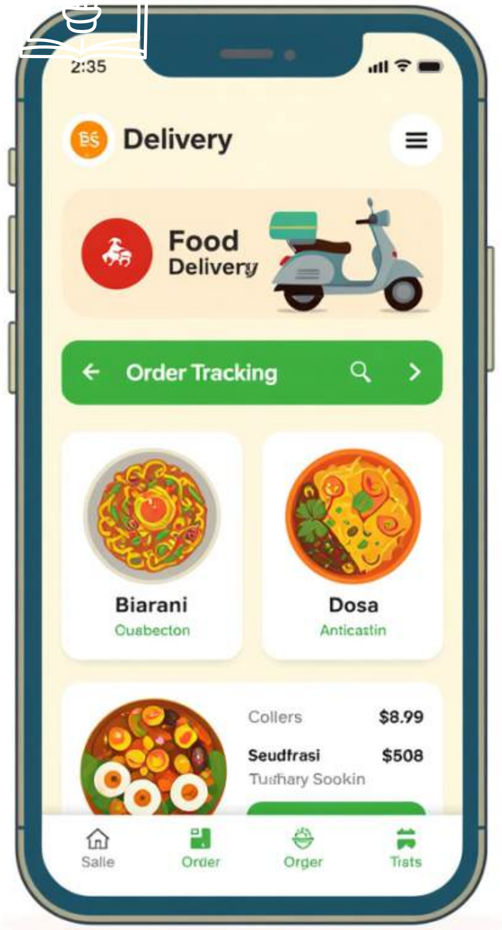
🌧️ LIVE DATA = FRESH INFO EVERY TIME!

WHAT IS AN API?

MEET THE WAITER!

API = WAITER!

You ask → Waiter brings your food (data!)
Just like a waiter takes your order to the kitchen and brings back your food, an API takes your request to a server and brings back the data you need!
Examples: Swiggy, Zomato use APIs every day!



API = APPLICATION PROGRAMMING INTERFACE — YOUR DATA DELIVERY HERO!

USEEFFECT –

THE SCHOOL BELL ALARM!

useEffect = School bell that rings once! It runs your code automatically when the page loads — just like a bell starting class without anyone pressing a button.

Empty array [] means "run only once" — like a bell that rings at the start of school and then stays quiet for the rest of the day!

SIMPLE USEEFFECT
CODE PREVIEW

```
useEffect(() => {
console.log("Page loaded!");
}, []);
```

The empty array [] means this code runs only once — right when the page appears. No repeats!

[] = ONCE EMPTY ARRAY MEANS THE CODE RUNS ONLY ONE TIME WHEN THE PAGE LOADS

AUTO RUN USEEFFECT RUNS AUTOMATICALLY — NO BUTTON NEEDED TO TRIGGER IT!

CODE-ALONG: DAILY

MOTIVATION CARD!

LET'S BUILD A FUN CARD WITH LIVE DATA!

We're going to create a card that shows a daily motivation quote and a fun fact about Indian superheroes and ISRO! We'll use fetch() and useEffect() to bring live data into our React app. Get ready to code along!

- WE'LL USE FETCH() AND USEEFFECT()
- API URL: API.PLANRS.IN/GRADE5/FUN-FACTS

BUILDING OUR LIVE DATA CARD STEP BY STEP!

START CODING!

01

02

06

FETCHING DATA

WITH USEEFFECT

```
useEffect(() => {
setLoading(true);
fetch('https://api.planrs.in
/grade5/fun-facts')
.then(res => res.json())
.then(data => {
setFunFact(data.fact);
setLoading(false);
});
}, []);
```

Fetch = call the waiter!

Ask the API to bring your data.

Show loading = patience indicator

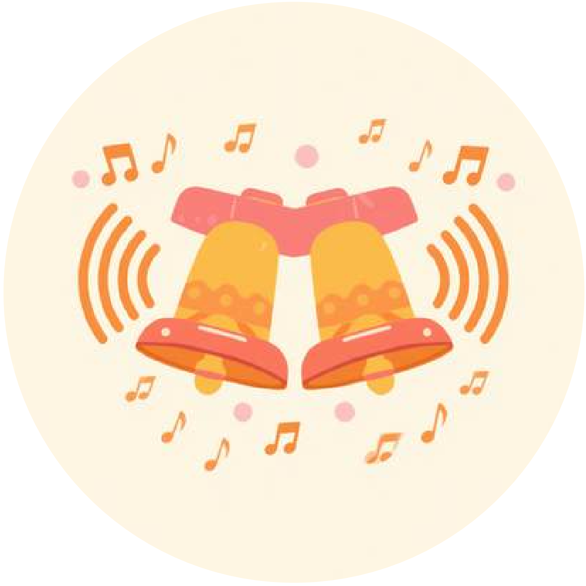
Tell users to wait while data loads.

Update card when data arrives

Display the fun fact once ready!

QUIZ & DEBUG
CHALLENGE!

CONFUSED KID



BELLS RINGING!

CODING SUPERHERO

QUIZ & DEBUG

What happens if we forget [] in useEffect?

A) Code runs once

B) Code runs many times ✓

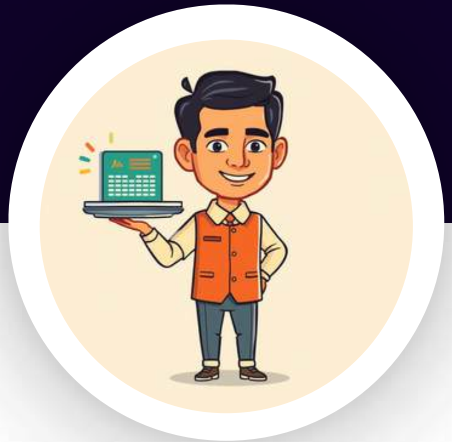
C) API waiter disappears

Bonus Challenge: Can you spot the API waiter in the code snippet?

You're a coding superhero in training!

RECAP & HOMEWORK MISSION!

useEffect = automatic helper (school bell)
Runs once when the page loads using the empty [ ] dependency array. Just like a school bell that rings automatically!



API = waiter bringing data
fetch() gets live info from the internet. Call the waiter, get your data delivered fresh every time!

Homework Mission 🏆
Build a weather badge for your favorite Indian city! Use useEffect + fetch() to show live weather data.

THANK YOU!

Great job, coding superheroes! 🚀

You learned about useEffect, APIs, and how to fetch live data with React.